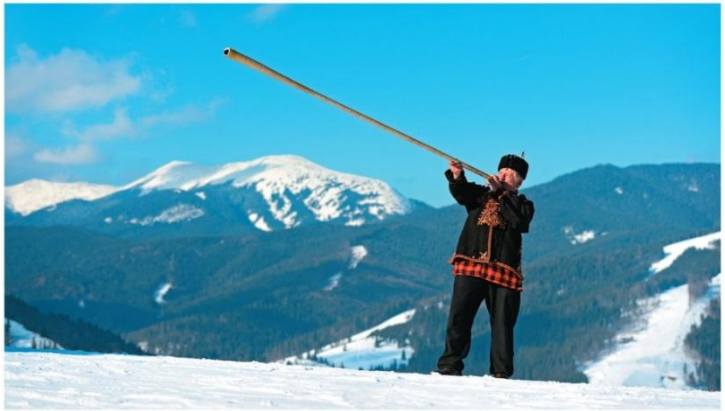
Learning Outcomes

In this chapter, you'll learn…

- how to describe a sound wave in terms of either particle displacements or pressure fluctuations.
- how to calculate the speed and intensity of sound waves.
- what determines the particular resonance frequencies of sound produced by an organ or a flute.
- what happens when sound waves from different sources overlap.
- the Doppler effect: Why the pitch of a siren changes as it moves past you.

Introduction

- The sound from a horn travels more slowly on a cold winter day high in the mountains than on a warm summer day at sea level.
- What determines the speed of sound in air?
- We can think of a sound wave either in terms of the displacement of the particles or of the pressure it exerts.
- How do humans actually perceive sound?
- Why is the frequency of sound from a moving source different from that of a stationary source?



Sound Waves

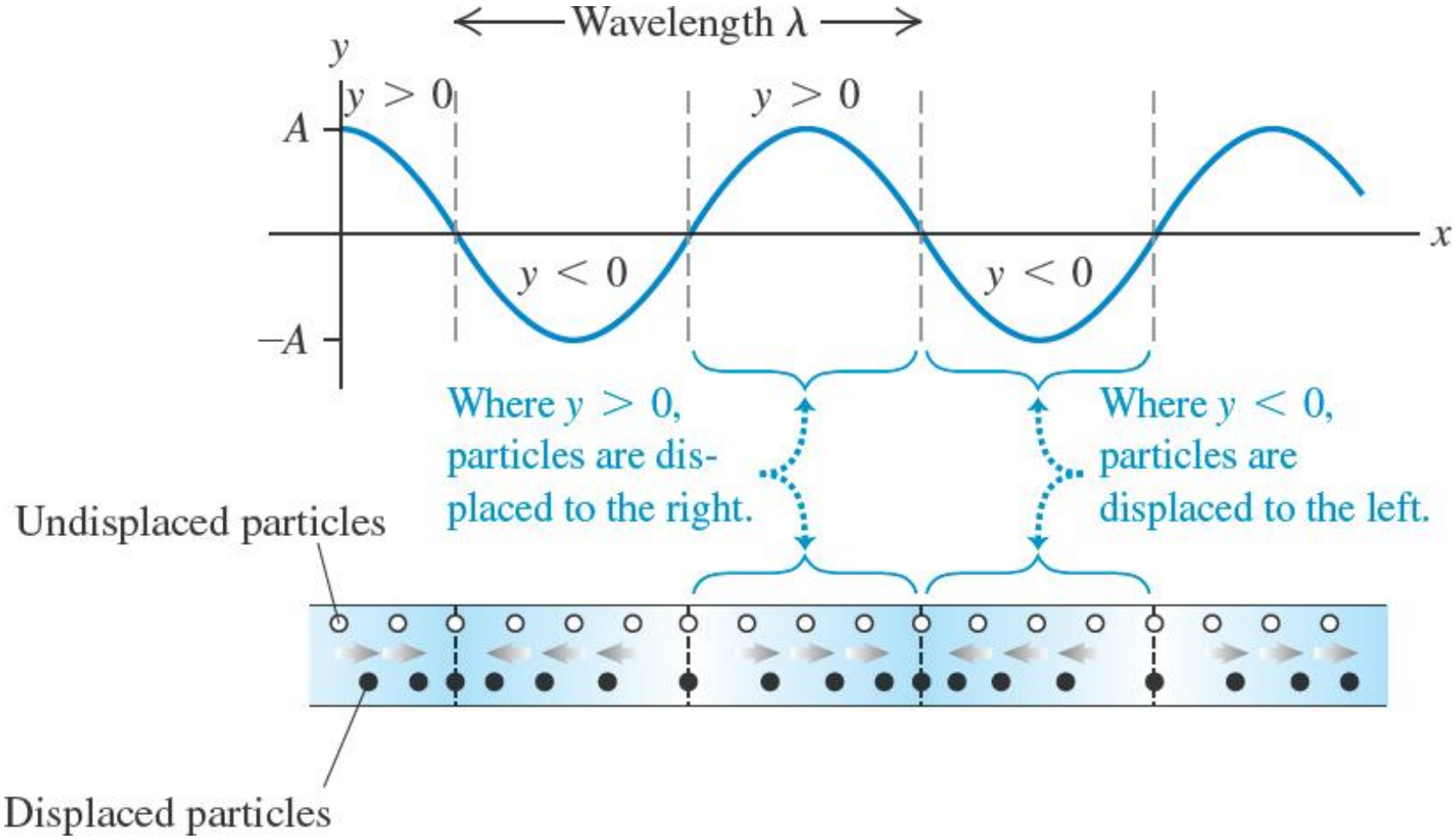
- **Sound** is simply any longitudinal wave in a medium.
- The **audible range** of frequency for humans is between about 20 Hz and 20,000 Hz.
- For a sinusoidal sound wave traveling in the *x*-direction, the wave function *y*(*x*, *t*) gives the instantaneous displacement *y* of a particle in the medium at position *x* and time *t*:

$$y(x,t) = A\cos(kx - \omega t)$$

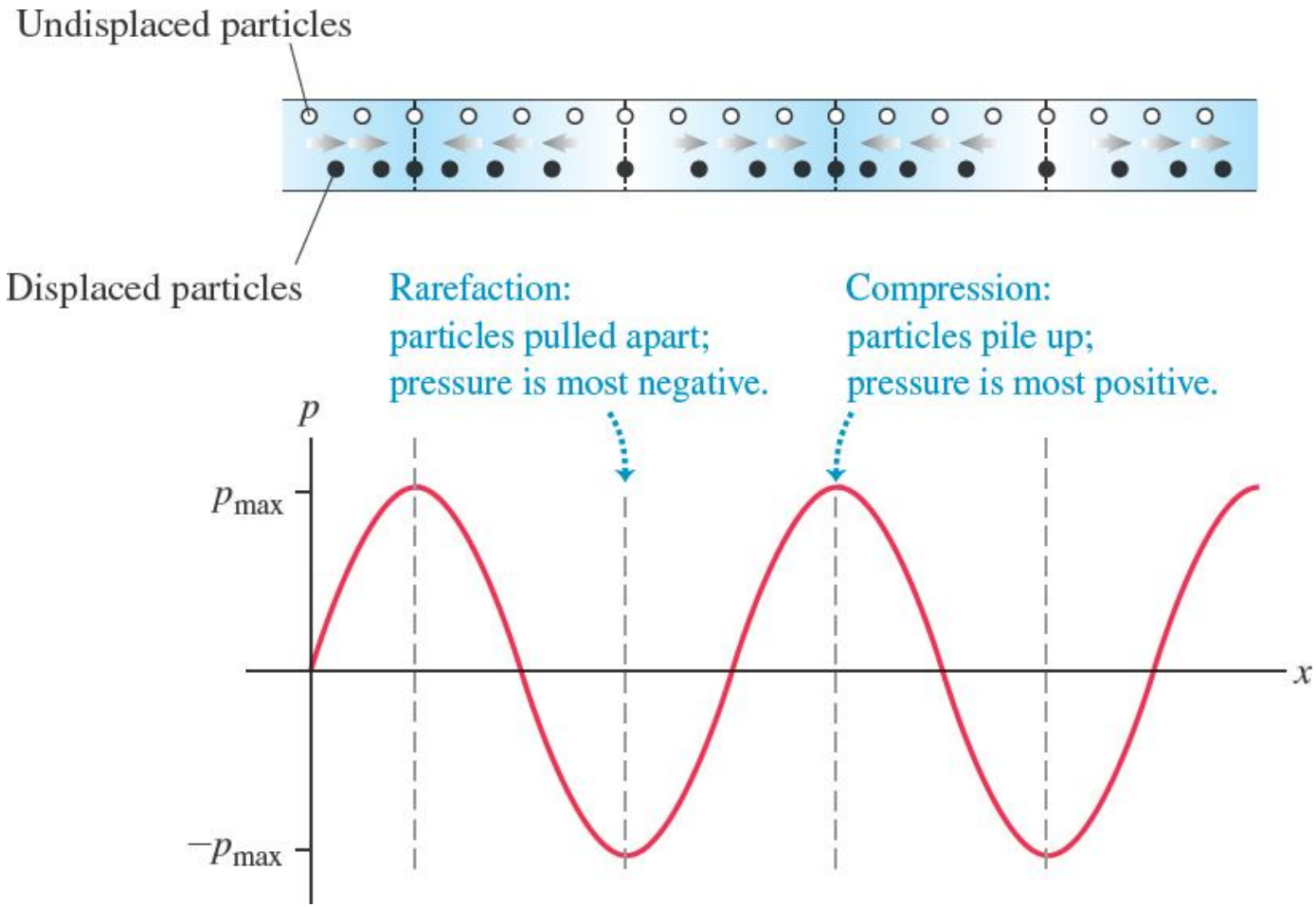
(sound wave propagating in the + x-direction)

- In a longitudinal wave the displacements are **parallel** to the direction of travel of the wave, so distances *x* and *y* are measured parallel to each other, not perpendicular as in a transverse wave.

Different Ways to Describe a Sound Wave (1 of 3)



Different Ways to Describe a Sound Wave (2 of 3)



Different Ways to Describe a Sound Wave (3 of 3)

- Sound can be described mathematically as a displacement wave:

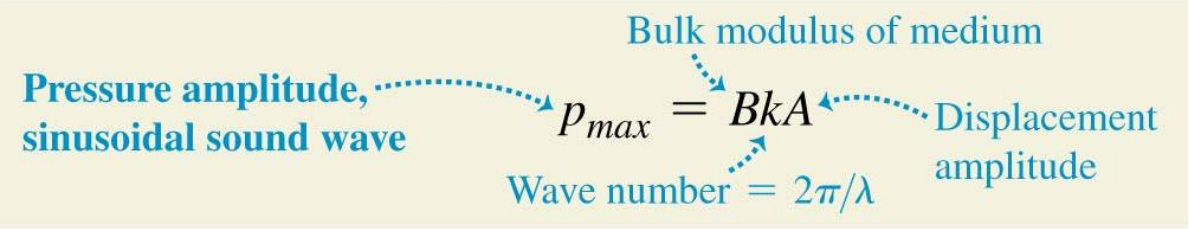
$$y(x,t) = A\cos(kx - \omega t)$$

(sound wave propagating in the + x-direction)

- The same sound wave can alternatively be described mathematically as a pressure wave:

$$p(x,t) = BkA\sin(kx - \omega t)$$

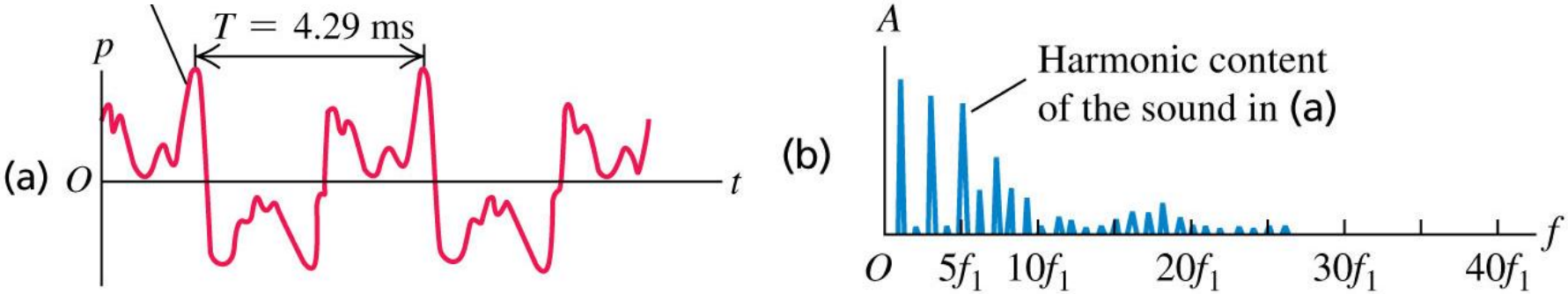
- The quantity *BkA* represents the maximum pressure fluctuation, and is called the **pressure amplitude**:



Ex. 1 Example 16.1 (Section 16.1) showed that for sound waves in air with frequency 1000 Hz, a displacement amplitude of 1.2×10^{-8} m produces a pressure amplitude of 3.0×10^{-2} Pa. a) What is the wavelength of these waves? b) For 1000-Hz waves in air, what displacement amplitude would be needed for the pressure amplitude to be at the pain threshold, which is 30 Pa? c) For what wavelength and frequency will waves with a displacement amplitude of 1.2×10^{-8} m produce a pressure amplitude of 1.5×10^{-3} Pa?

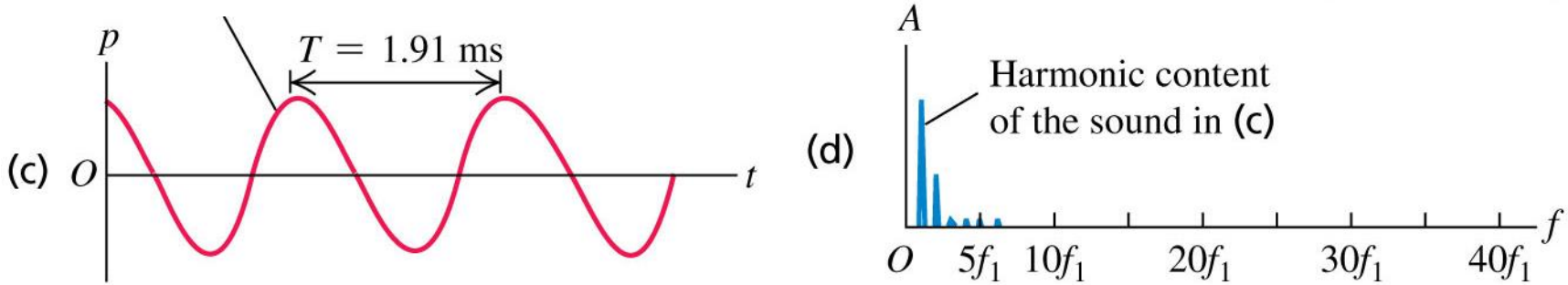
Perception of Sound Waves (1 of 2)

- Shown in (a) is the pressure fluctuation versus time for a clarinet with fundamental frequency f_1 = 233 Hz:
- The mathematical process of translating the pressure–time graph (a) into a graph of harmonic content (b) is called **Fourier analysis**.



Perception of Sound Waves (2 of 2)

- Shown in (c) is the pressure fluctuation versus time for an alto recorder with fundamental frequency f_1 = 523 Hz.
- The mathematical process of translating the pressure–time graph (c) into a graph of harmonic content (d) is called **Fourier analysis**.



Speed of Sound Waves

- The speed of sound depends on the characteristics of the medium.
- In a fluid, such as water, the speed of sound is:

Speed of a longitudinal wave in a fluid $v = \sqrt{\dfrac{B}{\rho}}$

B: Bulk modulus of fluid

ρ: Density of fluid

- In a solid rod or bar, the speed of sound is:

Speed of a longitudinal wave in a solid rod $v = \sqrt{\dfrac{Y}{\rho}}$

Y: Young's modulus of rod material

ρ: Density of rod material

- In an ideal gas, such as air, the speed of sound is:

Speed of sound in an ideal gas $v = \sqrt{\dfrac{\gamma R T}{M}}$

γ: Ratio of heat capacities

R: Gas constant

T: Absolute temperature

M: Molar mass

- Video Tutor Solution: Example 16.4

Table 16.1 Speed of Sound in Various Bulk Materials

Material	Speed of Sound (m/s)
Gases	
Air (20 degrees Celsius)	344
Helium (20 degrees Celsius)	999
Hydrogen (20 degrees Celsius)	1330
Liquids	
Liquid helium (4 K)	211
Mercury (20 degrees Celsius)	1451
Water (0 degrees Celsius)	1402
Water (20 degrees Celsius)	1482
Water (100 degrees Celsius)	1543
Solids	
Aluminum	6420
Lead	1960
Steel	5941

Ultrasonic Imaging

- This three-dimensional image of a fetus in the womb was made using a sequence of ultrasound scans.
- Each individual scan reveals a two-dimensional “slice” through the fetus; many such slices were then combined digitally.
- Ultrasound imaging is also used to study heart valve action and to detect tumors.

Sound Intensity

- Sound waves transfer energy from one region of space to another.
- The **wave intensity** *I* is defined as the time average rate at which wave energy is transported per unit area across a surface perpendicular to the direction of propagation.
- The intensity of a sinusoidal sound wave is proportional to the square of the pressure amplitude:

Intensity of a sinusoidal sound wave in a fluid

$$I = \frac{p_{max}^2}{2\rho v} = \frac{p_{max}^2}{2\sqrt{\rho B}}$$

Pressure amplitude

Wave speed

Density of fluid

Bulk modulus of fluid

The Decibel Scale

- Because the ear is sensitive over a broad range of intensities, a logarithmic measure of intensity called **sound intensity level** is often used:

Intensity of sound wave

Sound intensity level

$$\beta = (10\ \text{dB}) \log \frac{I}{I_0}$$

Reference intensity $= 10^{-12}\ \text{W/m}^2$

Logarithm to base 10

- The chosen reference intensity I_0 is approximately the threshold of human hearing at 1000 Hz.
- Sound intensity levels are expressed in **decibels**, abbreviated as dB.
- Video Tutor Solution: Example 16.8

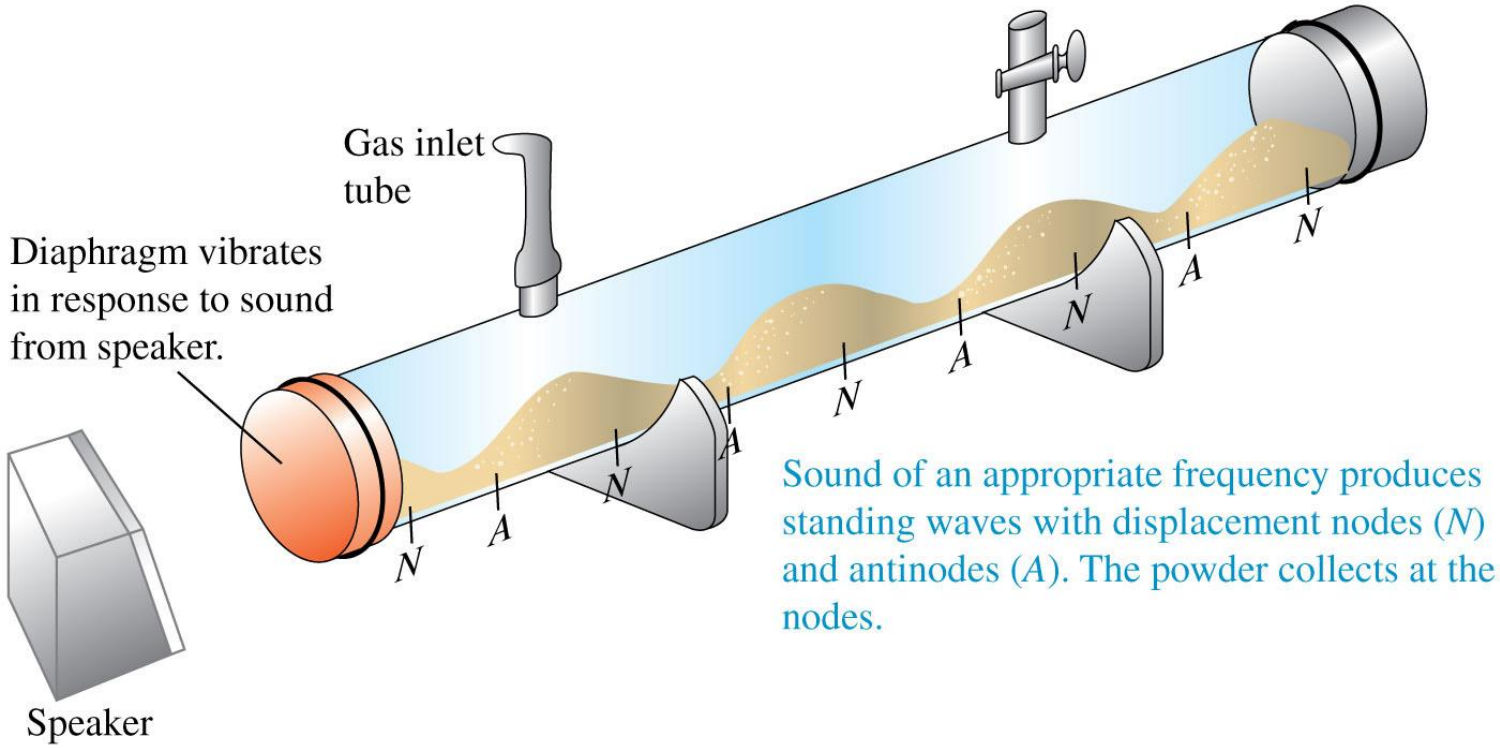
Ex. 15 A sound wave in air at 20°C has a frequency of 320 Hz and a displacement amplitude of 5.00×10^{-3} mm. For this sound wave calculate the (a) pressure amplitude (in Pa); (b) intensity (in W/m^2); (c) sound intensity level (in decibels).

Sound Intensity Levels – Representative Values

Source or Description of Sound	Sound Intensity Level, β (dB)	Intensity, I (W/m^2)
Military jet aircraft 30 m away	140	10^2
Threshold of pain	120	10^0
Elevated train	90	10^{-3}
Busy street traffic	70	10^{-5}
Quiet radio in home	40	10^{-8}
Average whisper	20	10^{-10}
Threshold of hearing at 1000 Hz	0	10^{-12}

Standing Sound Waves

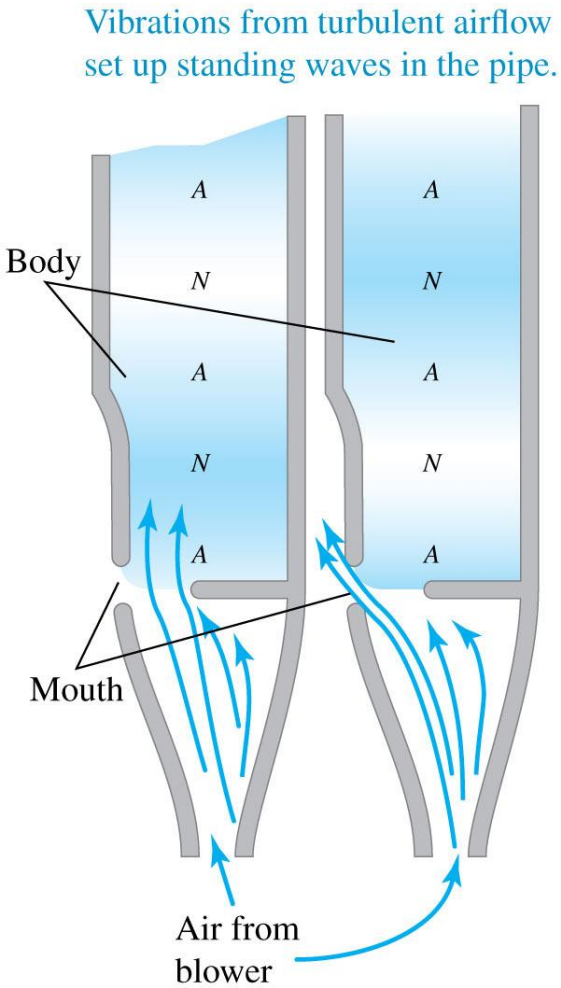
- When sound waves propagate in a fluid in a pipe, the waves are reflected from the ends and a standing wave can be produced.



Sound of an appropriate frequency produces standing waves with displacement nodes (N) and antinodes (A). The powder collects at the nodes.

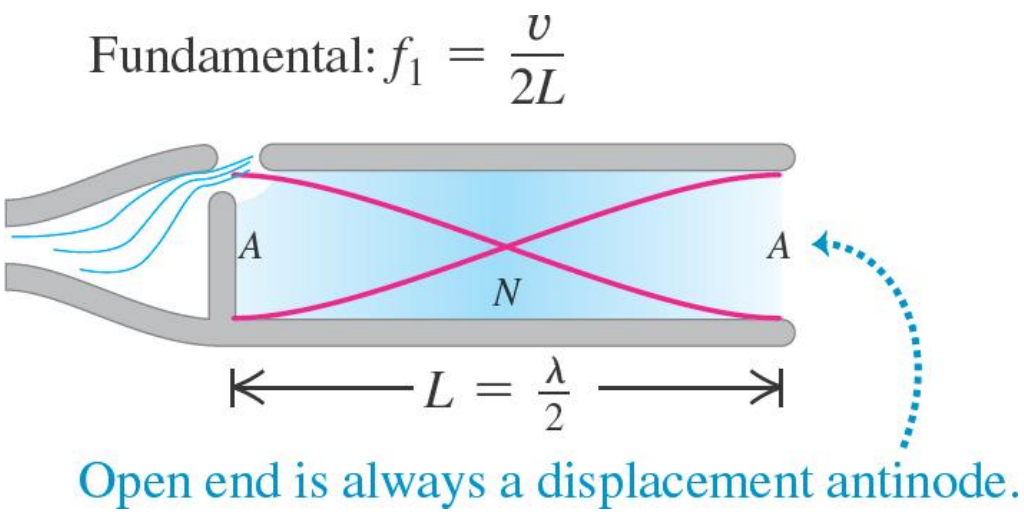
Organ Pipes

- Organ pipes of different sizes produce tones with different frequencies (bottom figure).
- The figure at the right shows displacement nodes in two cross-sections of an organ pipe at two instants that are one-half period apart. The blue shading shows pressure variation.



Harmonics in an Open Pipe (1 of 3)

- The fundamental frequency of an open pipe is shown.
- The shading indicates the pressure variations.
- The red curves are graphs of the displacement along the pipe axis at two instants separated in time by one half-period.



Harmonics in an Open Pipe (2 of 3)

- Higher harmonics in an open pipe have frequency:

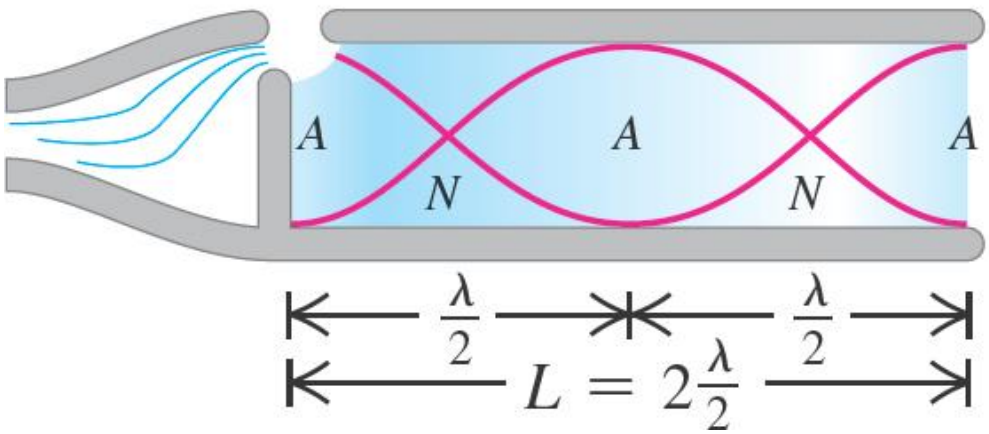
Frequency of nth harmonic ($n = 1, 2, 3, \ldots$)

$$f_n = \frac{nv}{2L}$$

v: Speed of sound in pipe

L: Length of pipe

$$\text{Second harmonic: } f_2 = 2\frac{v}{2L} = 2f_1$$



Harmonics in an Open Pipe (3 of 3)

- Higher harmonics in an open pipe have frequency:

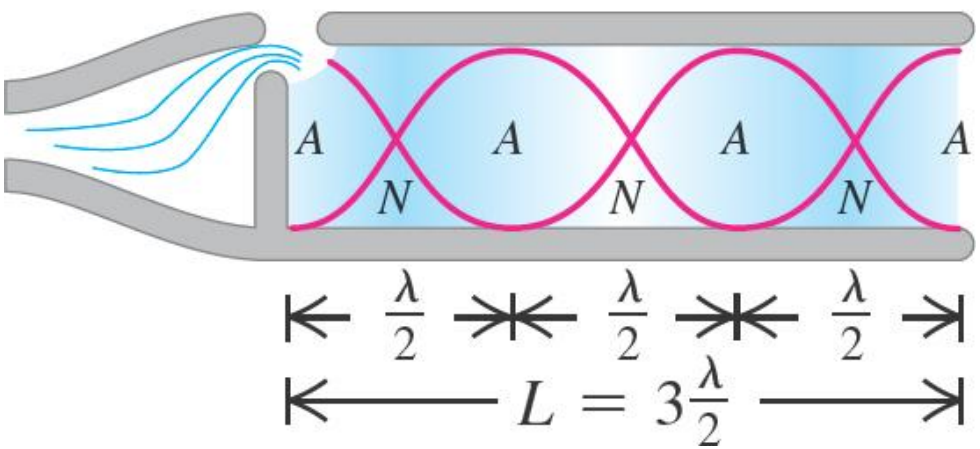
Frequency of nth harmonic ($n = 1, 2, 3, \ldots$)

$$f_n = \frac{nv}{2L}$$

Speed of sound in pipe

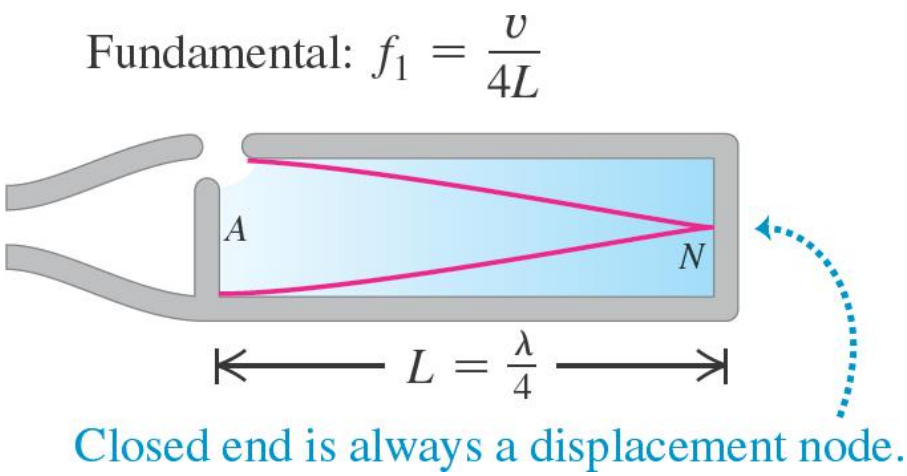
Length of pipe

$$\text{Third harmonic: } f_3 = 3\frac{v}{2L} = 3f_1$$



Harmonics in a Stopped Pipe (1 of 3)

- The fundamental frequency of a stopped pipe is shown.
- The shading indicates the pressure variations.
- The red curves are graphs of the displacement along the pipe axis at two instants separated in time by one half-period.



- Video Tutor Solution: Example 16.12

Harmonics in a Stopped Pipe (2 of 3)

- Higher harmonics in a stopped pipe have frequency:

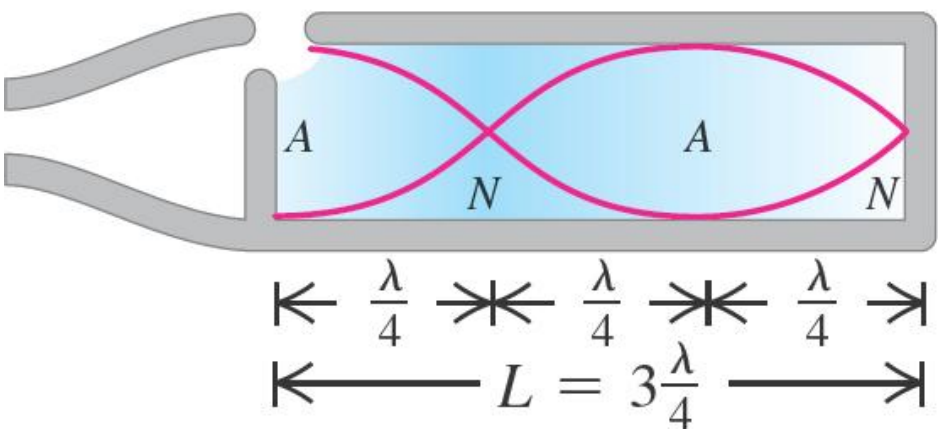
Frequency of nth harmonic ($n = 1, 3, 5, \ldots$)

$$f_n = \frac{nv}{4L}$$

v: Speed of sound in pipe

L: Length of pipe

$$\text{Third harmonic: } f_3 = 3\frac{v}{4L} = 3f_1$$



Harmonics in a Stopped Pipe (3 of 3)

- Higher harmonics in a stopped pipe have frequency:

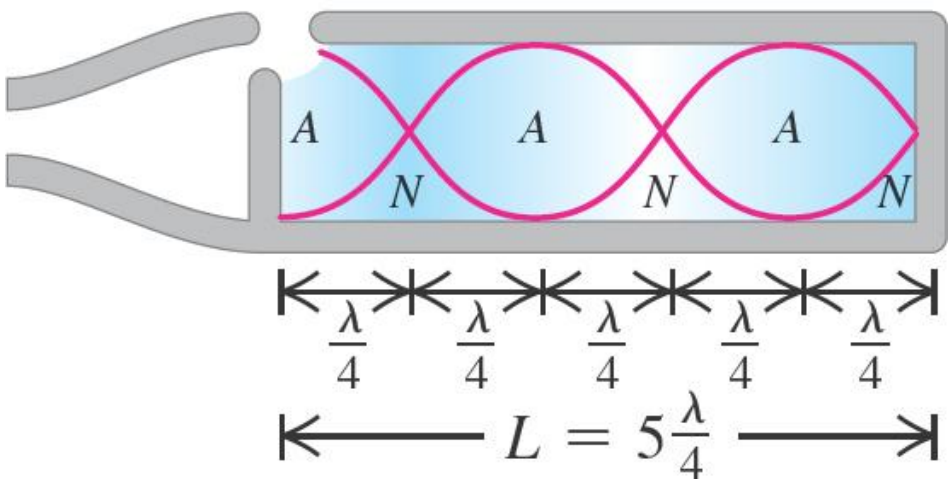
Frequency of nth harmonic ($n = 1, 3, 5, \ldots$)

$$f_n = \frac{nv}{4L}$$

v: Speed of sound in pipe

L: Length of pipe

$$\text{Fifth harmonic: } f_5 = 5\frac{v}{4L} = 5f_1$$



Ex. 27 The longest pipe found in most medium-size pipe organs is 4.88 m (16 ft) long. What is the frequency of the note corresponding to the fundamental mode if the pipe is (a) open at both ends. (b) open at one end and close at the other?

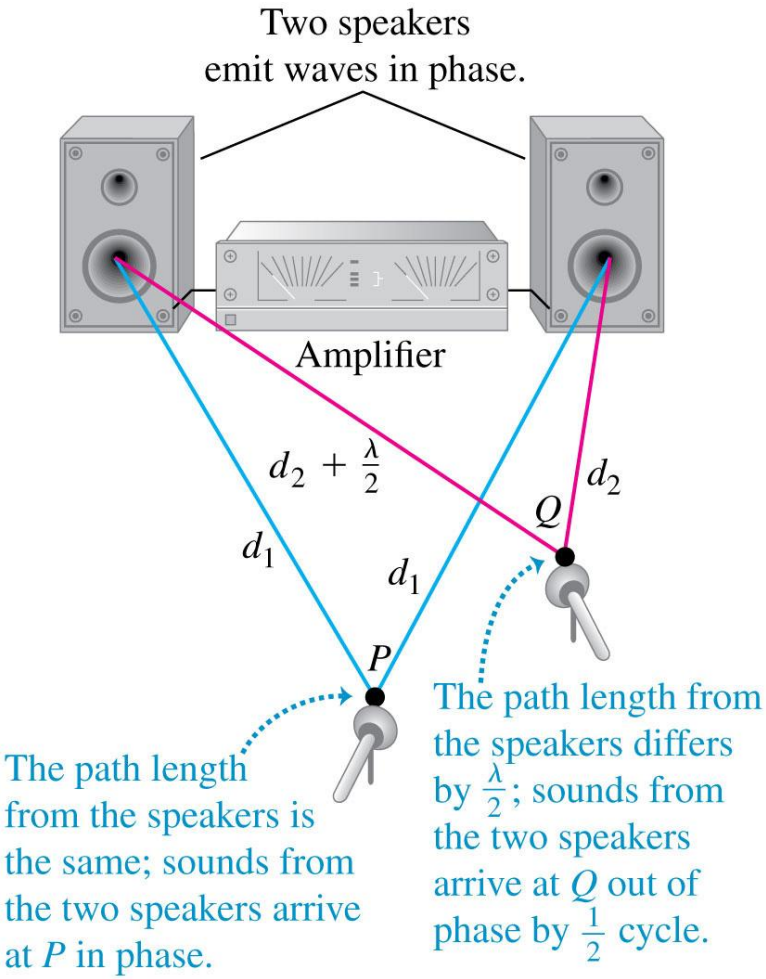
Resonance and Sound

- The frequency of the sound from this trumpet exactly matches one of the normal-mode frequencies of the goblet.
- The resonant vibrations of the goblet have such large amplitude that the goblet tears itself apart.

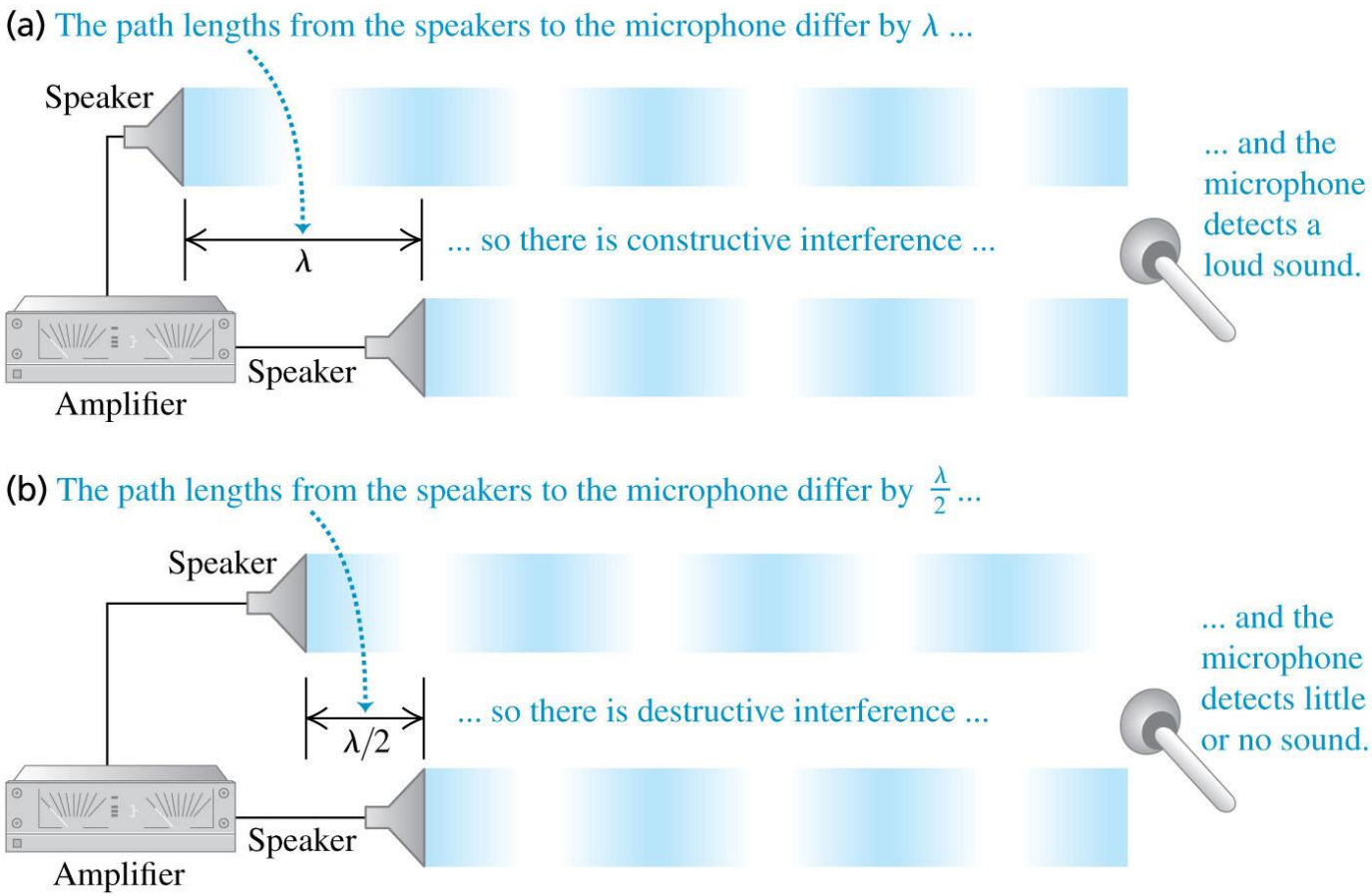
Ex. 30 You have a stopped pipe of adjustable length close to a taut 62.0-cm, 7.25-g wire under a tension of 4110 N. You want to adjust the length of the pipe so that, when it produces sound at its fundamental frequency, this sound causes the wire to vibrate in its second *overtone* with very large amplitude. How long should the pipe be?

Interference (1 of 2)

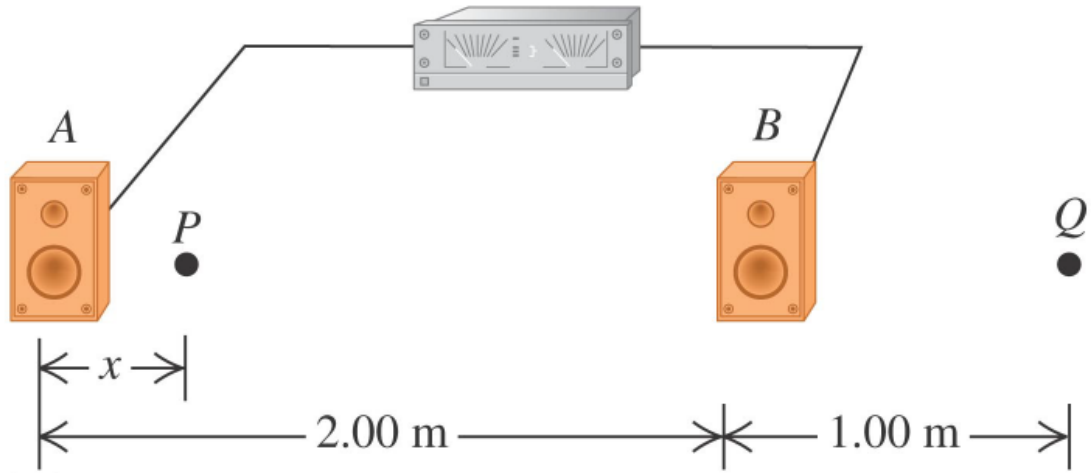
- When two or more waves overlap in the same region of space they **interfere**.
- In the figure, two speakers are driven by the same amplifier.
- Constructive interference occurs at point *P*, and destructive interference occurs at point *Q*.



Interference (2 of 2)



Ex. 33 Two loudspeakers, A and B (Fig. E16.35), are driven by the same amplifier and emit sinusoidal waves in phase. Speaker B is 2.00 m to the right of speaker A. Consider point Q along the extension of the line connecting the speakers, 1.0 m to the right of speaker B. Both speakers emit sound waves that travel directly from the speaker to point Q. What is the lowest frequency for which (a) *constructive* interference occurs at point Q; (b) *destructive* interference occurs at point Q?



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Beats (1 of 2)

- **Beats** are heard when two tones of slightly different frequency (f_a and f_b) are sounded together.
- The amplitude rises and falls with a new, much lower, frequency called the **beat frequency**:

Beat frequency for waves a and b

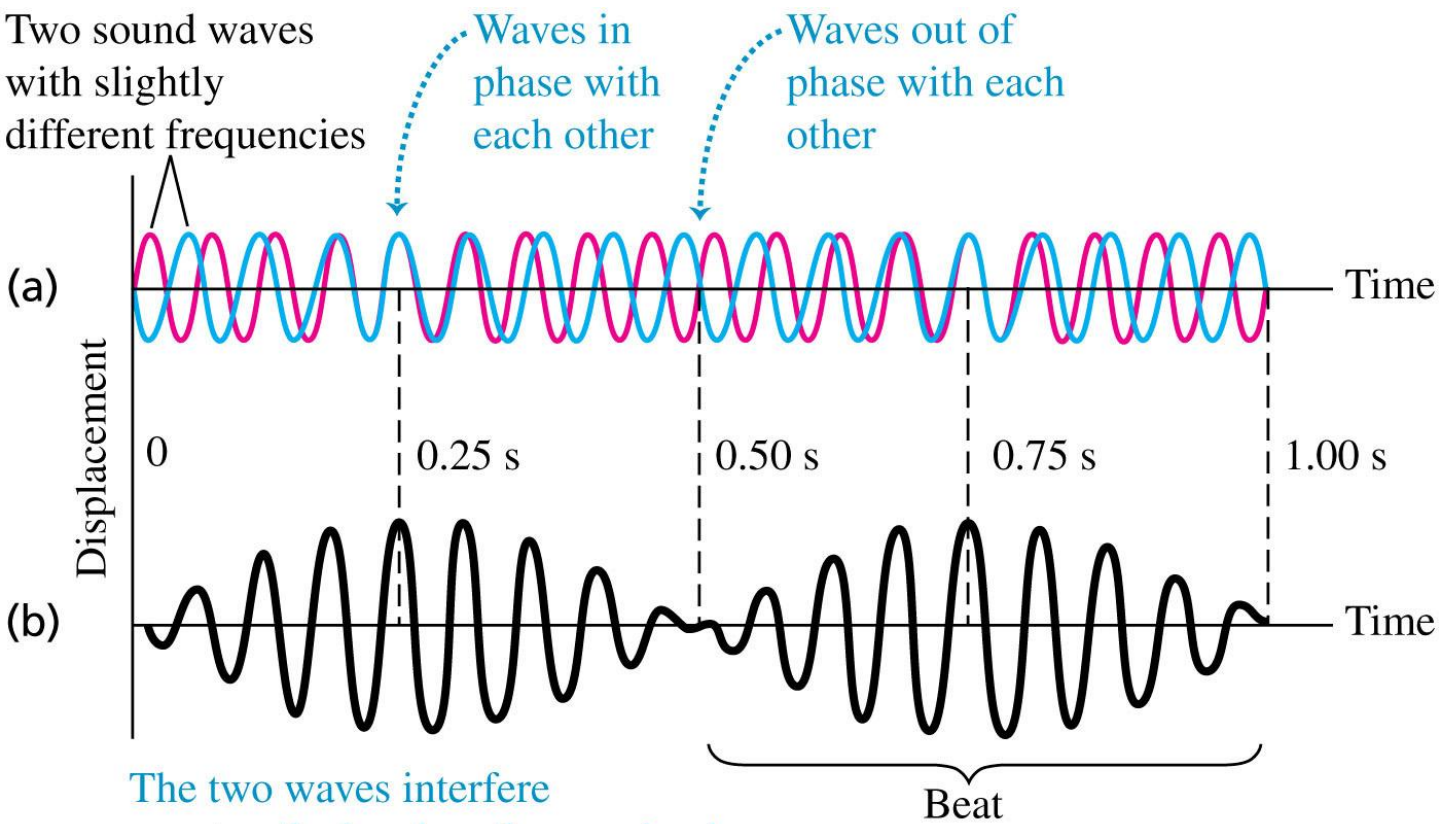
$$f_{\text{beat}} = f_a - f_b$$

Frequency of wave a

Frequency of wave b (lower than f_a)

- Beats between two tones can be heard up to a maximum beat frequency of about 6 or 7 Hz.
- Listening for beats is an important technique in tuning all musical instruments.

Beats (2 of 2)



The two waves interfere constructively when they are in phase and destructively when they are a half-cycle out of phase. The resultant wave rises and falls in intensity, forming beats.

Ex 40: Two organ pipes, open at one end but closed at the other, are each 1.14 m long. One is now lengthened by 2.00 cm. Find the beat frequency they produce when playing together in their fundamentals.

The Doppler Effect (1 of 2)

- The **Doppler effect** for sound is the shift in frequency when there is motion of the source of sound *S*, the listener *L*, or both:

Frequency heard by listener

Velocity of listener (+ if from L toward S, − if opposite)

$$f_L = \frac{v + v_L}{v + v_S} f_S$$

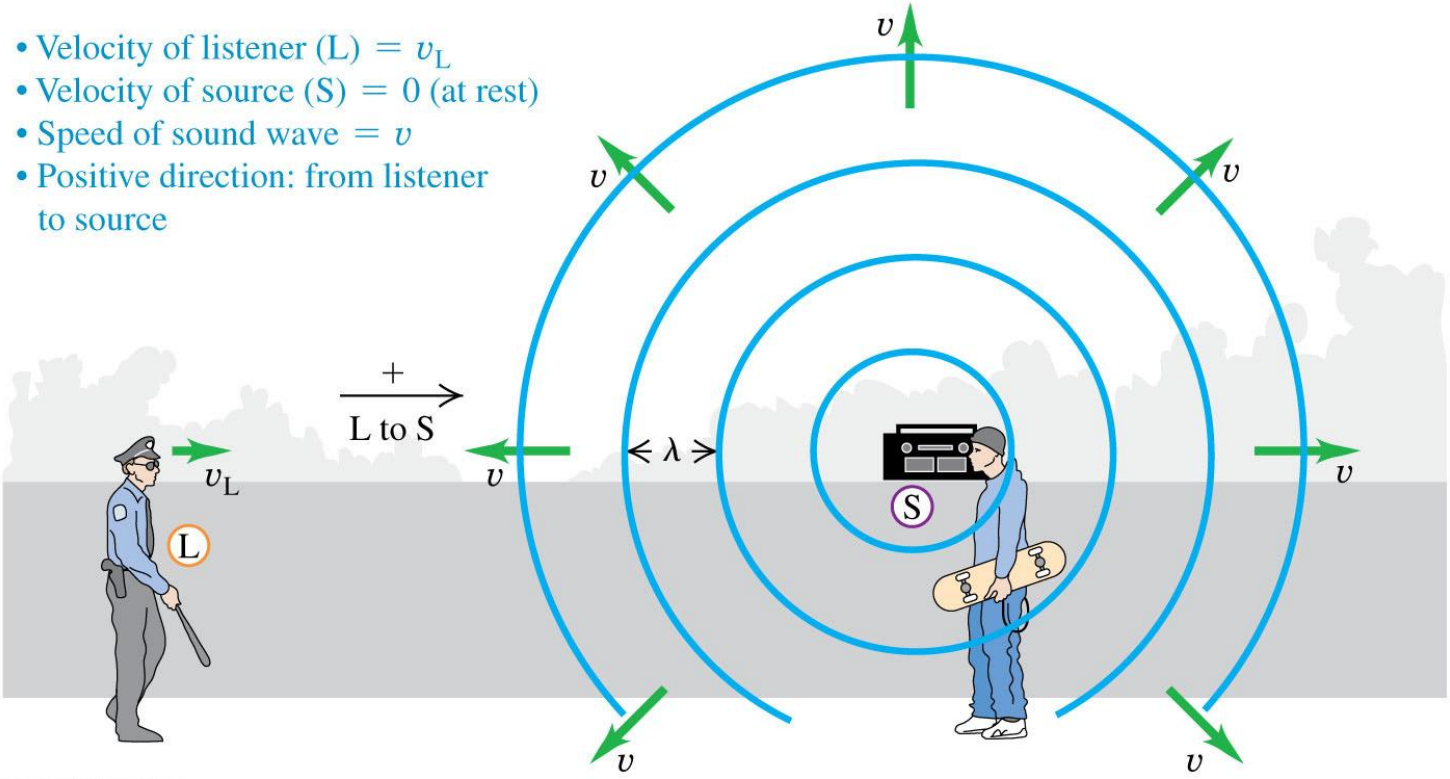
Frequency emitted by source

Speed of sound

Velocity of source (+ if from L toward S, − if opposite)

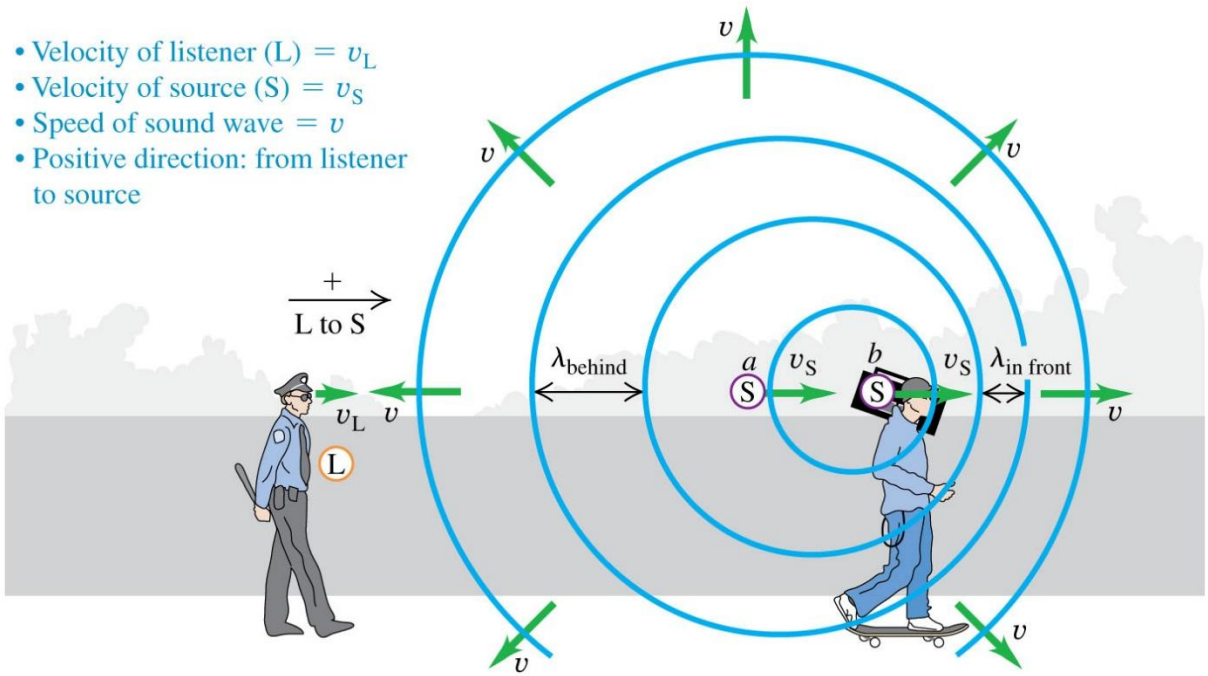
The Doppler Effect: Moving Listener

- A listener moving **toward** a stationary source hears a frequency that is **higher** than the source frequency.



The Doppler Effect: Moving Source

- When a source is moving **away** from a listener, the waves behind the source are stretched to a **longer** wavelength.



The Doppler Effect (2 of 2)

- The Doppler effect explains the observed change in pitch of the siren on a fire engine or ambulance.
- The frequency is high ($f_L > f_S$) when it is approaching you ($u_S < 0$).
- The frequency is low ($f_L < f_S$) when it is moving away from you ($u_S > 0$).

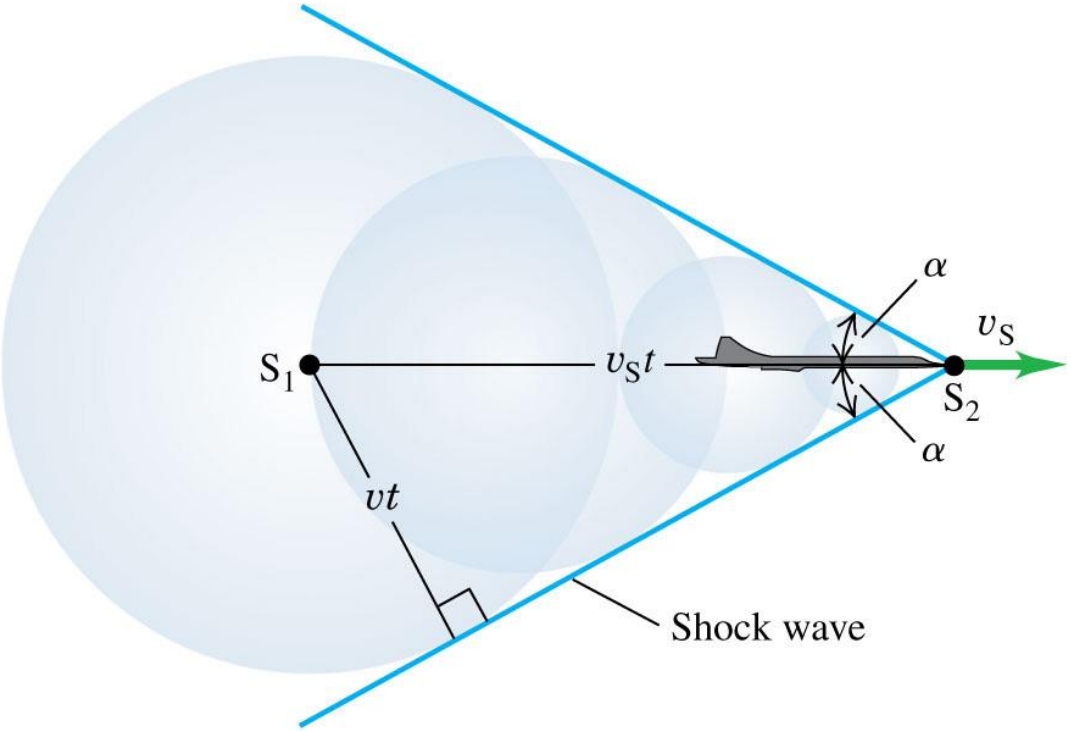
- Video Tutor Solution: Example 16.17

Ex. 44 Moving Source vs. Moving listener. a) A sound source producing 1.00 kHz waves moves toward a stationary listener at $\frac{1}{2}$ the speed of sound. What frequency will the listener hear? b) Suppose instead that the source is stationary and the listener moves toward the source at $\frac{1}{2}$ the speed of sound. What frequency does the listener hear? How does your answer compare to that in part (a)? Explain on physical grounds why the two answers differ.

Shock Waves

- When an object travels at a speed greater than the speed of sound in air, it creates a shock wave.

$$\sin \alpha = \frac{v}{v_s}$$



Ex. 53 A jet plane flies overhead at Mach 1.70 and at a constant altitude of 1250 m. a) What is the angle α of the shock-wave cone? b) How much time after the plane passes directly overhead do you hear the sonic boom? Neglect the variation of the speed of sound with altitude.